

Taher Batterywala

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Education

Vellore Institute of Technology

B.Tech in CSE (Specialization in AI/ML) | CGPA: 8.30

Bhopal, India

Sep 2023 – Jun 2027

S.I.C.A. Senior Secondary School

Senior Secondary Education (CBSE, Class I-XII)

Indore, India

Apr 2008 – Mar 2023

Technical Skills

Programming Languages	C++, Python, Java, SQL, HTML/CSS, JavaScript
CS Fundamentals	Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks
AI & Machine Learning	Deep Learning, Computer Vision, Generative AI, Large Language Models (LLMs), NLP, Retrieval-Augmented Generation (RAG)
Frameworks & Tools	TensorFlow, Scikit-learn, LangChain, Pandas, Flask, Streamlit, Amazon Web Services (AWS), Git/GitHub, Render, REST APIs

Experience

FOSSEE, IIT Bombay | Summer Intern | [GitHub](#) | [Merged PR](#) | [Certificate](#)

May 2026 – Jul 2026

- Implemented a subprocess isolation layer utilizing Base64-serialized I/O pipes for a local AI Copilot, resolving C++ memory segmentation faults and eliminating UI thread latency (from 1.5s to near-zero).
- Integrated an offline RAG pipeline using Qwen2.5:3b and MiniCPM-V, tokenizing and indexing engineering manuals into 74 discrete ChromaDB vector database layers for local retrieval.
- Wrote a regex-driven garbage collection routine to eliminate file-system memory leaks from temporary clipboard images, and built a self-healing config script to resolve startup crashes on fresh installations.

Projects

Tirakot AI: Offline Voice-Activated Desktop Assistant | [GitHub](#)

Local AI & Systems Engineering Project | Tech: Python, PyTorch, Ollama, Whisper, CustomTkinter, asyncio

- Developed a fully offline, voice-activated desktop AI assistant, orchestrating three concurrent Machine Learning models (Qwen Large Language Model, Whisper Speech-to-Text, Edge Text-to-Speech) under a strict 2.34 GB Video RAM budget.
- Built eight real-time Operating System (OS) automation tools—including application launching and a Retrieval-Augmented Generation (RAG) web search—triggered by intercepting LLM JSON tool-calling via first-token stream interception.
- Implemented an Exponential Moving Average (EMA) noise floor and Voice Activity Detection (VAD) gating to eliminate speech hallucination errors, integrated seamlessly into a custom Tkinter Graphical User Interface (GUI).

GenWealth: AI Portfolio Manager & Financial Advisor | [GitHub](#)

Machine Learning & Financial Forecasting Project | Tech: PyTorch, Stable-Baselines3 (PPO), Scikit-learn, MongoDB (RAG), Groq/Gemini

- Built an automated portfolio manager that uses a hybrid PyTorch LSTM and Random Forest engine to forecast stock trends, paired with a Deep Reinforcement Learning (PPO) agent to automatically size and execute positions.
- Integrated a conversational financial advisor into the system using a Multi-Agent LLM pipeline and a MongoDB Vector RAG database, allowing users to query live market news and receive hallucination-free analysis reports.
- Engineered a zero-lookahead simulation engine with dynamic profit-trimming and stop-loss rules, demonstrating superior risk-adjusted returns and significantly reducing maximum portfolio drawdown across an 8-year historical backtest.

GamerSense: Behavioral Segmentation Engine | [GitHub](#)

Data Science & ML Engineering Project | Tech: Python, Scikit-learn, XGBoost, SMOTE, Flask, Pandas

- Engineered an end-to-end ML pipeline with dataclass configurations, deploying K-Means clustering to segment 3,024 mobile players into actionable behavioral personas.
- Benchmarked 7 algorithms utilizing SMOTE on a severe 38:6:1 imbalanced dataset, achieving 77.2% classification accuracy with an SVC to predict in-app spending segments.

Leadership & Extracurriculars

Google Developer Groups VIT Bhopal | Core Team Member | [Certificate](#)

Nov 2024 – Jun 2025

Achievements & Certifications

- Competitive Programming** – Solved 400+ Data Structures and Algorithms problems on LeetCode using C++ [Profile](#)
- Introduction to Internet of Things (Topper 5%)** – NPTEL [Certificate](#)
- Applied Machine Learning in Python** – University of Michigan (Coursera) [Certificate](#)
- Cloud Computing** – NPTEL (Swayam) [Certificate](#)